Ray Tracing Attention

Shared causal BVHs and lazy state materialization

Branch A query

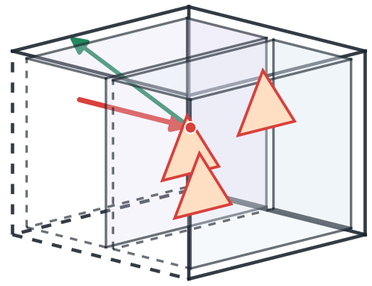
Branch B query

Causal identity

Model / geometry namespace

Complete optical dependencies

Shared history BVH

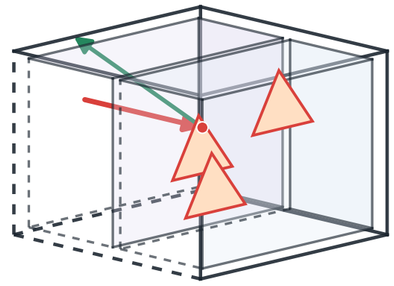


Boundary marker

Shared block references

High-D row / optical address

Causal address BVH



Token + successor

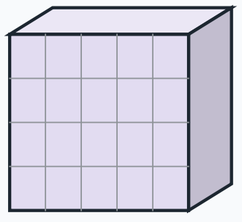
$(y_t,\ \mathcal{Q}_{t+1})$

Input ends / output ends

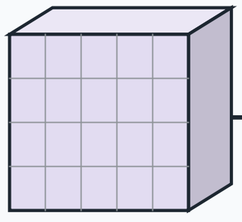
Common roots / intermediate boundaries

Shared across conversation branches

Shared storage and propagation



B_0



B_1

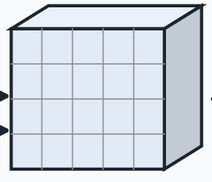
Immutable light packets + BLAS

Branch A references

$[B_0, B_1, B_2]$

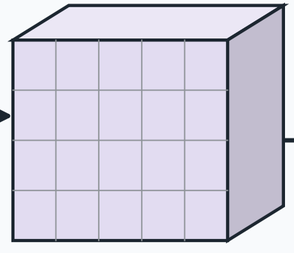
Branch B references

$[B_0, B_1, B_3]$



h_p

High-D boundary row



S_{live}

One live recurrent field

Address miss

Optical propagation

Materialize on address miss

Update current field

$$M_{\text{shared}} = \sum_{B \in \mathcal{U}} |B| + 4D\,H + O(H)$$